

Classification of Celestial Bodies using SVM

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Abstract. This paper presents an end-to-end machine learning pipeline for classifying astrophysical objects in SDSS images which were in the form of FITS files. With the combination of an SVM classifier, advanced feature engineering and selection, the model scored a balanced-test-set accuracy of 74.6% and an overall accuracy of 65.6% on the entire dataset which is a substantial 17.17% higher than a baseline model. The study has pointed out the effectiveness of the SVM in the differentiation of astronomical classes in addition to the discovery of morphological ambiguity between distant galaxies, quasars and stars as one of the major challenges. This confusion resulted in tradeoffs in recall and precision, specifically, in galaxies and stars. The results highlight the significance of systematic methods that include feature engineering, data balancing, and hyperparameter optimization and indicate that future developments might involve the combination of multi-band photometry and spectroscopic data to eliminate the limitations that are inherent in morphology-based classification.

Keywords: Support Vector Machines (SVM), Celestial Object Classification, Sloan Digital Sky Survey (SDSS), Feature Engineering.

1 Introduction

For generations, the vast sky has grabbed the attention of human curiosity, compelling us to understand the mysteries beyond the skies. But the advances in modern technology has allowed us to obtain large volumes of data about stars, galaxies, and other celestial phenomena [3]. Though the large amount of data is helpful, it renders manual classification of this data an impossible task [3], [14], [15]. This results in the need of an automated system to accurately classify these images, which will help astronomers to speed up the process of classifying celestial objects [4], [14]. We obtain millions of images on galaxies, stars and quasars through astronomical surveys [1], [3]. Although spectroscopic methods studying the light emitted from celestial objects to study what they are can be used to identify these objects, they are costly and cannot be scaled to such large datasets [15]. Shape and color are among the most fundamental features that are referred when classifying celestial objects [5], [7], [13], but noise in images makes this task challenging. The classification of celestial bodies is a fundamental problem in astronomy, and traditional methods such as manual inspection or spectral analysis are inefficient for the massive image datasets produced by modern telescopes [3]. In this study, we use image processing to provide a practical

alternative, where simple features like shape and color can capture key differences between stars, galaxies, and quasars [5], [8], [12]. Previous approaches like manual inspection and spectral analysis often emphasize brightness or spectra, which are not always reliable under observational limitations. In this study, we focus on extracting shape and color features from astronomical images and applying Support Vector Machines (SVM) for classification [4], [6], [7]. We hypothesize that this combination of basic image analysis with SVM can provide an efficient, interpretable, and accurate method for large-scale celestial classification [6]. To address these issues, we developed a full machine learning pipeline for the classification of celestial objects. We designed a pipeline, starting with background subtraction and ending with object segmentation that aims to process the images in exactly the same way. With stars being very hard to distinguish from distant galaxies, we did not stop at simple shapes and have derived 61 different features, such as texture analysis and radial light profiles, which helps the model tell them apart. We also addressed the issue of uneven class distribution by resampling our training set and changing the SVM parameters to get better performance on minority classes. **The key contributions of this work are summarized as follows:**

1. An end-to-end SVM-based pipeline for classifying celestial objects from SDSS FITS images using systematic preprocessing and segmentation.
2. A 61-dimensional multi-domain feature set combining morphological, photometric, texture, and structural profile descriptors to address morphological ambiguity.
3. Comprehensive evaluation using balanced and real-world imbalanced datasets with precision, recall, F1-score, confusion matrices, and baseline comparisons.
4. Critical analysis of intrinsic limitations of single-band morphological classification and discussion of practical extensions using multi-band photometry and spectroscopy.

2 Literature Survey

Modern sky surveys produce millions of images of stars, galaxies, and quasars [3], [14], [15]. Analyzing all these images by hand is very time-consuming and requires a lot of effort. To make this process faster, researchers use image processing and machine learning techniques [3]. Simple shape-based features, such as size and elongation, are useful for distinguishing different types of celestial bodies [8], [13]. This makes automated classification an important area of research. However, there are still some challenges. Stars, galaxies, and quasars sometimes look very similar, which makes them hard to classify correctly [1]. Images can also have noise, which can reduce the accuracy of feature extraction. Deep learning methods, like convolutional neural networks (CNNs), can give high accuracy but need powerful computers and large datasets [2]. For small projects or limited resources, these methods may not be practical. Different approaches have been tried in the past. Early studies used statistical and machine learning methods to classify stars and galaxies [3]. Richards et al. [1] showed that supervised learning on SDSS data performs better than manual classification. Later, CNNs became popular because they can automatically learn features from images [2]. But CNNs need a lot of computing power. Simpler methods, such as Support Vector Machines (SVMs), work well with shape features and are easier to run, while still giving good accuracy [4], [6], [7].

Table 1. Literature survey comparison.

Study	Focus	Method	Strengths	Weakness
[1]	Star-galaxy separation	ML on survey data	Better than manual methods	Did not cover quasars
[2]	Galaxy shapes	CNN	Very high accuracy	High computation cost
[3]	SDSS classification	SVM, DT	Better than humans	Sensitive to noise

In 2024 and 2025, more recent advances have also studied this trade-off between model complexity and efficiency. Wu et al. [16] showed that ensemble learning such as Random Forest may be more effective than complicated neural networks for the classification of SDSS Objects using only photometric features close to a perfect scale. A 2025 study on SDSS Data Release 18 [17] also demonstrated that gradient boosting methods can achieve an accuracy higher than 98% using spectral features of objects, indicating that for tabular data classical machine learning is still highly effective. However, the field is also moving in the direction of "lightweight" models for image-based tasks. As emphasized by Li et al. [18], it is incredibly important that efficient algorithms can be used, which can run on limited hardware, without requiring the massive computational overhead of standard deep learning. This can be supported by the recent studies on such architectures, such as EfficientNet-G3 [19], which strives to limit the number of parameters without lowering the classification rate. Such philosophy is in line with our work, which shows that a well tuned SVM can be used as a strong, low-resource substitute of morphological classification.

Over time, methods have evolved from manual checks and basic statistics to machine learning models like SVMs and Decision Trees [3], [4], [14], [15], and more recently to deep learning approaches [2]. Deep learning gives high accuracy but requires more resources. Lightweight methods like SVMs are still useful for projects with limited hardware or small datasets [6], [7]. There is still a need for approaches that are accurate yet simple and resource-efficient. Our study focuses on using shape features with an SVM classifier [6], [8]. This method is practical for student-level projects and gives a good balance between accuracy and simplicity.

3 Methodology

Our approach utilizes a systematic pipeline in categorization of the celestial objects: obtaining data related to a professional astronomical survey, preprocessing the images, dividing them into objects, multistage feature extractions and designs, and finally training and classifying with the help of the SVM model. All these are executed through automated scripts so that the entire process of raw images up to final classification is similar and reproducible.

3.1 Data Acquisition

The data were retrieved in SDSS programmatically using astroquery library. A script query was used to search a 3-arcminute radius of random sky positions iteratively using the sky positions of objects spectroscopically defined as a GALAXY, STAR, or QSO. It downloaded single-band (r-band) image and metadata of the image in the FITS format in case of positive match. This was carried out in a 1,000 images production, the unprocessed information in this project. The distribution of the entire

imbalanced dataset (1,000 samples) is as follows: GALAXY: 507 samples (50.7% of the data). QSO: 271 samples (27.1% of the dataset). STAR: 222 samples (22.2% of the dataset). To have an objective model training, a balanced subset was selected by under sampling the majority classes so that the relationship is similar to the smallest one. This was followed by a Balanced Training Set of 532 samples (around 177-178 samples per class) and a Balanced Test Set of 134 samples (around 44-45 samples per class) which were finally validated.

3.2 Preprocessing

The preprocessing step helps to improve the image fidelity to be analyzed. The FITS files were loaded one by one and NaN pixel values which might be caused by sensor artifacts were changed to zero. A median filter was applied on the entire image so as to isolate the target object against the sky background which tends to have noise or faint distant sources. This process essentially removes the large-scale background structure. Image contrast was then adjusted by normalizing the pixel intensities to the image range specified by the 1st and the 99th percentile of the image, which made sure the extreme outliers were not influencing the segmentation process.

3.3 Object Segmentation

Segmentation removes the primary celestial object in the preprocessed picture. A dynamic Gaussian thresholding method was applied, that calculated a distinct threshold on the different parts of the picture. It is thus unresponsive to different levels of local backgrounds. The binary mask obtained was then smoothed with morphological operators: an opening operator was used to remove tiny noise pixels and other spurious objects, after which a closing operator was used to seal tiny holes or gaps within the central object. The last step involved labeling the connected components and only the biggest single component in terms of pixel area was left to take the shape of the definitive mask of the object of interest.

3.4 Feature Extraction and Engineering

After segmenting the primary object, a 35-component feature vector was generated to describe it. This feature vector had four types of features:

1. **Morphological and Photometric Features:** Properties including aspect ratio, solidity, extent, morphological and photometric features of the object were computed using the mask of the object. The photometric characteristics like mean, maximum and minimum pixel intensity in the mask were also calculated.
2. **Texture Features:** Local Binary Patterns were employed to measure the amount of the texture of the object surface. The image became a 10-bin normalized histogram which in turn summarized the image after LBP transformation.
3. **Shape-Invariant Features:** The seven Hu Moments extracted were the contour of the object and they are not only translation-invariant, scale-invariant and rotation-invariant. The raw moment values were log transformed to bring about numerical stability.
4. **Structural Profile Features:** A radial profile of the object based on the mean pixel intensity in eight concentric and equidistant rings or bins about the centroid

of the object was calculated. This describes the distribution of the object brightness of its center to the periphery.

These 35 raw features were used to build a bigger feature vector with 61 components in order to enhance the discrimination capabilities of the model. More interaction features, such as intensity ratios, measures of compactness, log transformed area, and statistical descriptors of the radial profile bins, were generated by doing so. Mathematically derived features included:

Intensity Interactions:

- intensity range (Max - Min)
- intensity contrast (Max - Mean / Mean)
- intensity normalized (Mean / Max)

Advanced Morphological Descriptors:

- compactness, circularity, shape complexity, and elongation.

Combined Features:

- area intensity (Area $\times$ Mean Intensity)
- area perimeter ratio

Logarithmic Transforms:

- Log-scaled versions of Area, Mean Intensity, and Max Intensity (to handle heavy-tailed distributions)

Radial Profile Statistics:

- Gradients between rings (7 features), Mean, Standard Deviation, and Max/Min ratio of the profile

Statistical Aggregates:

- Mean and Standard Deviation of the Hu Moments and LBP texture distributions

The 50 most informative features were chosen through a supervised filter-based method to optimize the model performance and efficiency. Following an initial extraction of 35 base features encompassing morphological descriptors, photometric statistics, and structural profiles, the feature space was expanded to 61 dimensions through explicit engineering of geometric ratios, intensity interactions, and statistical aggregates. SelectKBest algorithm was used thereafter, to rank each attribute according to its Mutual Information (`mutual_info_classif`) score against the target celestial classes. This measure was selected due to its capability to represent complicated, non-linear correlations among astronomical features and category of objects that is required to resolve the natural morphological ambiguities between far galaxies, stars and quasars. By retaining only, the top 50 features with the highest information gain, the pipeline successfully reduced computational complexity and minimized model variance, avoiding overfitting at the expense of maintaining much of the important discriminatory capability needed to get large-scale classification accuracy.

4 Results and Discussion

This section is a full quantitative performance analysis of the optimized final classifier. The model is based on a Support Vector Machine (SVM), which is an effective algorithm in the high-dimensional feature spaces and is able to learn complex, non-linear boundaries because it makes use of the kernel trick.

The classification pipeline operates on a set of numerical features extracted from the raw FITS images. Each image was loaded using a preprocessing script which, as described in the Methodology, involved a background subtraction step followed by a segmentation of the main celestial object. From this object segmentation, 35 base features were obtained which measure its morphological, intensity, texture (LBP), and radial profile characteristics. These base features were further feature engineered to create a larger set of 61 features by including intensity ratios, logarithmic transforms and interaction terms.

This set of 61 features is not processed by a plain SVM, but through a complete scikit-learn Pipeline, which is described in the Methodology. The pipeline constitutes the whole process of transforming and classifying the data in a uniform and a repeatable way. The optimized pipeline is comprised of the following stages:

- **Feature Selection:** SelectKBest using the `mutual_info_classif` score, which automatically picks the top 50 most informative features of the 61 engineered features.
- **Data Scaling:** StandardScaler, that normalizes the features by cutting off the mean and scaling them to unit variance. This step is required to get the best results with SVM.
- **Classifier Description:** A Support Vector Classifier (SVC) with a Radial Basis Function (RBF) kernel was used. Hyperparameter optimization determined the most effective parameters as $C = 10$ (regularization parameter) and $\gamma = 0.05$ (kernel coefficient). To address class imbalance in the training data, the `class_weight=balanced` option was applied.

To obtain the objective knowledge of the distinguishing characteristics of each celestial class, the classifier has been trained on a balanced dataset (532 samples). The trained model was then tested on three datasets to give a detailed assessment of its learning capacity, generalization capacity and robustness:

- **Balanced Training Set (532 samples):** This was used to test the basic learning ability of the model, as well as, serve as a baseline in the diagnosis of possible overfitting.
- **Balanced Test Set (134 samples):** This test set was employed to test the actual generalization of the classifier on previously not seen but class balanced data.
- **Full Imbalanced Dataset (1000 samples):** This dataset was used to test the classifier under realistic, in the field conditions, where the GALAXY class is dominant (50.7% of the samples). This evaluation measures the robustness of the model to naturally occurring class distributions.

4.1 Overall Accuracy

Overall accuracy provides a general assessment of how accurately the model was found to be, which is the percentage of all predictions that were accurate. This metric is employed to measure the overall efficiency of the model and to diagnose overfitting. The

balanced test set gave the model a 74.63% overall accuracy. This outcome is considered a key indicator of the prediction strength of the model on unknown data. Figure 1 presents a visual comparison of the overall accuracy across all three datasets.

The training set (74.44%) and the test set (74.63%) results are almost identical. Such a small difference of 0.19% (with test accuracy being slightly higher than training accuracy) is a strong indication that the model generalizes well to unseen data and is not overfitting. As expected, the accuracy on the full, unbalanced dataset declined to 65.60%. This drop of 9% compared to the balanced test

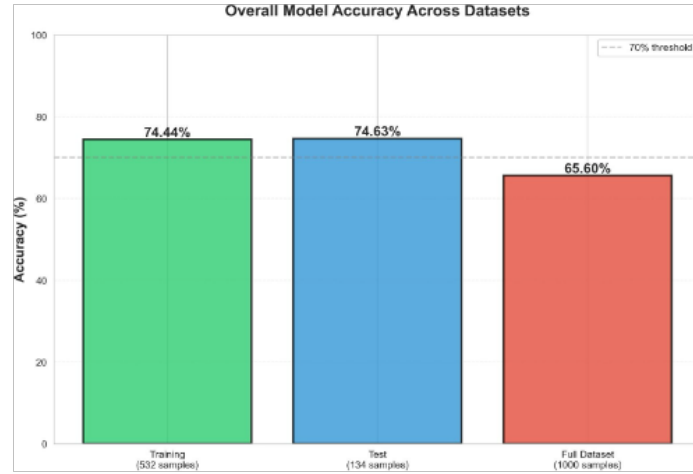


Fig. 1 Overall Model Accuracy Comparison.

set indicates that the situation of imbalance of classes in the real-world is indeed a major challenge, with the performance of the model being heavily affected by the GALAXY majority class. Although the peak accuracy of the final model of 74.63% on the balanced test set was an impressive 17-point improvement on the baseline, it failed to achieve the lofty goal of 80% accuracy. This analysis indicates that it is probably not a failure of the SVM model per se, but maybe a fundamental limitation due to the feature set. This is attributed to the fact that most of the distant galaxies are small and faint, and they form the shape and color characteristics that are very vague and are similar to the features of stars. In the same way, the quasars are also quasi-stellar and in many cases, cannot be distinguished by just morphological characteristics. To obtain accuracy greater than 75%, one would probably need to introduce more data, including spectroscopic measurements, which were beyond the means of this project.

4.2 Per-Class Performance Analysis

Overall accuracy is an effective summary, but in unbalanced situations it is misleading. To determine the specific strengths and weakness of the model in identifying any of the three types of celestial objects, a more detailed analysis per-class is necessary to comprehend the strengths and weaknesses of the model.

Precision and Recall: The two basic measures, which have more information about the model performance as compared to accuracy, are Precision and Recall. Precision is used to measure the accuracy of positive prediction. It provides the answer to the following

question:” Among all those things which the model predicted to be a STAR, what was the percentage of those things which were?” A high precision is that of low false-positive. The recall (or Sensitivity or Per-Class Accuracy) is used to measure how many instances of a class the model detects as true. It provides the answer to the question:” What is the proportion of things that the model found successfully among actual STARS in the data?” A large recall rate implies that there is a small false-negative rate. Figure 3 shows the trade-offs between the precision and recall of each class in the various datasets. The best outcomes are those of the complete, skewed dataset (red bars) which indicate that there are two clear and conflicting behaviors:

- **GALAXY:** The model has a high precision of about 80% but a low recall of 57.4 only. This shows that the model is conservative in estimating the prediction of the model is right four out of five times. However, it fails to identify (i.e., it “misses”) nearly 43% of all true galaxies in the dataset.
- **STAR:** The model shows the opposite problem, a very high recall of approximately 76% but a very low precision of only 50.9%. This means the model successfully finds over three-quarters of all true STARS, but it does so by “overpredicting” this class, resulting in nearly half of its STAR predictions being incorrect (i.e., they were actually GALAXYS or QSOs). This low precision for STARS is directly linked to the low recall for GALAXYS. The root cause is the high level of ambiguity between the classes. As will be detailed in the confusion matrix analysis (Section IV.C), the model frequently misclassifies distant, faint GALAXYS (which appear as small, “star-like” objects based on their shape features) as STARS. This single error pattern simultaneously drops the GALAXY recall (a “miss”) and destroys the STAR precision (a “false positive”).

F1-Score: The F1-Score is the harmonic mean of Precision and Recall. It offers a single score that balances both metrics. This score is especially helpful for assessing model performance on imbalanced datasets, where one metric can appear misleadingly high. Table II shows the detailed classification report for the balanced test set. In this balanced scenario, the model reaches a strong macro-average F1-Score of 0.746. The STAR class has the largest F1-Score at 0.791 indicating that it is the most reliably and accurately classified type of object when the distribution of classes is not a factor.

Table 2. Per-class classification report (balanced test set).

Class	Precision	Recall	F1-Score	Support
GALAXY	0.667	0.773	0.716	44
QSO	0.811	0.667	0.732	45
STAR	0.783	0.800	0.791	45
Macro Avg	0.754	0.747	0.746	134
Overall Acc	74.63%			134

4.3 Per-Class Accuracy (Recall):

As described, the Recall is identical to Per-Class Accuracy. This metric is presented in Figure 2 and it explains the percentage of the model explaining true instances of each of the classes. The results of the precision-recall analysis are supported by the chart. The STAR class was the most reliable on the test set, with the accuracy of 80.0%. Conversely, GALAXY class suffered a significant decrease in performance on

the full dataset with the accuracy decreasing to 57.4% on the full dataset as compared to 77.3% on the test set. The QSO was the most consistent in its performance and was not observed to vary significantly in all three datasets. This gap in performance specifically the low GALAXY recall on the full dataset is directly related to the problem of morphological ambiguity. As can be seen in the confusion matrix analysis, the model has often confused 133 true GALAXYs as STARS, on top of confusing 83 true GALAXYs as QSOs. This is due to the fact that beyond the limited distance, most of the galaxies are small and faint and have diffuse shape and color descriptions with stars and quasars.

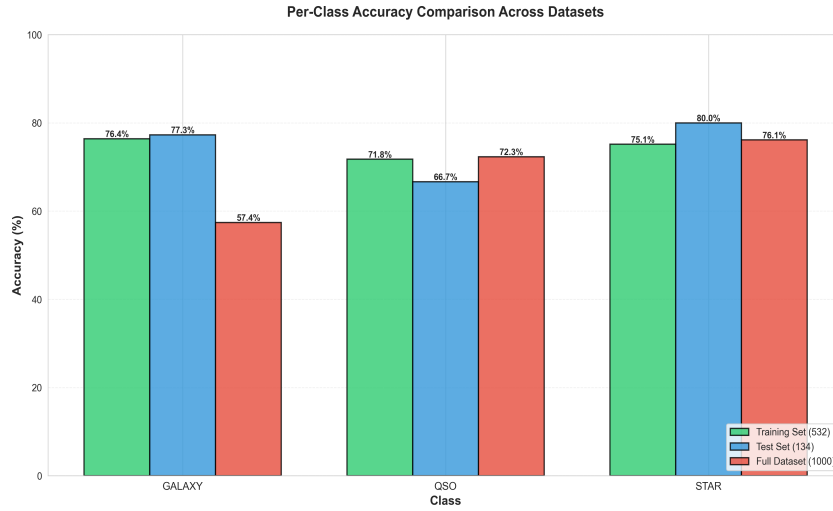


Fig. 2 Per-Class Accuracy (Recall) Comparison.

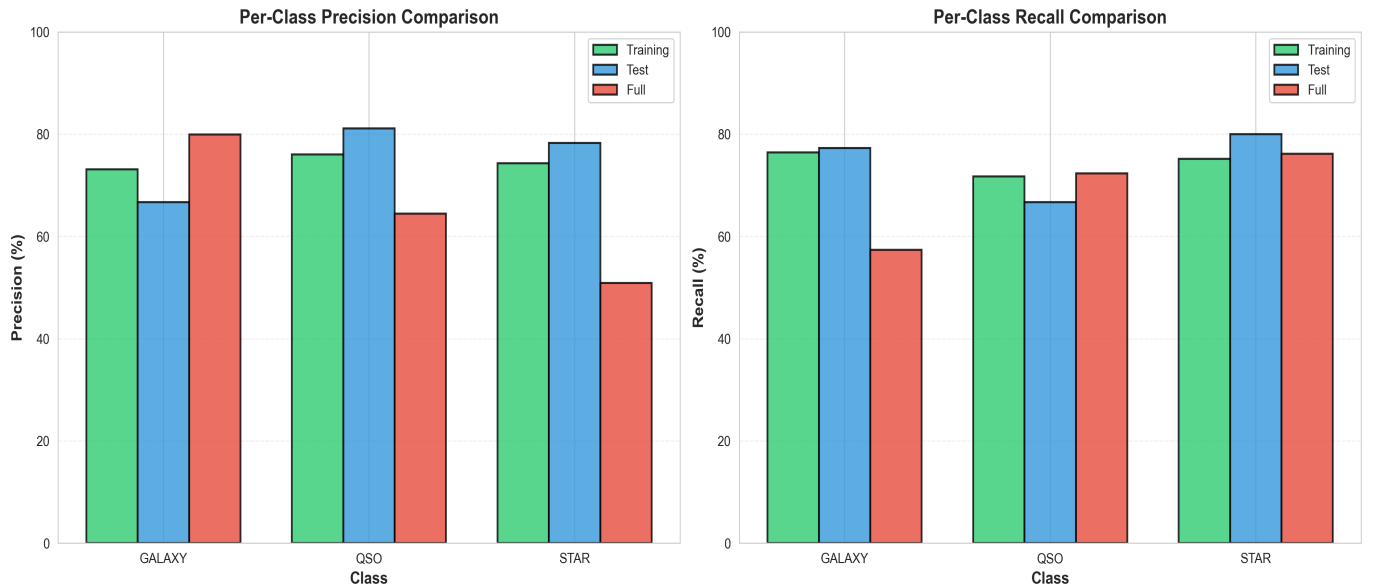


Fig. 3 Per-Class Precision (left) and Recall (right) Comparison.

4.4 Detailed Classification Analysis (Confusion Matrices)

In order to examine the particular error patterns and comprehend the manner in which the model exhibited systematic misclassification patterns, confusion matrices were formed. A confusion matrix gives a finer-grained break down of the correct predictions (diagonal), and incorrect predictions (off diagonal) by each class.

1. **Generalization Performance (Balanced Test Set):** Figure 4.A indicates the performance of the model on the balanced test set.
 - **Correct:** The model was able to recognize 34 GALAXYS, 30 QSOs, and 36 STARS correctly.
 - **Errors:** On the primary list of confusion (12 instances) was a wrong classification of true QSOs into GALAXYs which is indicative of a large overlap between the engineered characteristics of these two classes. One error pattern that occurred in secondary error was the classification of 7 true GALAXY as STARS.

Table 3. Per-class performance on full imbalanced dataset (1000 samples).

Class	Precision	Recall	F1-Score	Support
GALAXY	0.799	0.574	0.668	507
QSO	0.645	0.723	0.682	271
STAR	0.509	0.761	0.610	222
Macro Avg	0.651	0.686	0.653	1000
Overall Accuracy			65.60%	1000

2. **Real-World Performance (Full Imbalanced Dataset):** Figure 4.B shows how significant the issue of imbalance in the classes is on the predictions of the model.
 - **Correct:** The diagonal shows 291 GALAXYs, 196 QSOs, and 169 STARS were correctly identified.
 - **Errors:** The matrix clearly visualizes the trade-offs seen in the precision-recall chart
 - **GALAXY Recall:** The reason behind such a low recall of 57.4% of GALAXYS is explained by the fact that 133 true GALAXYs were mis-classified as STARS and 83 true GALAXYs were misclassified as QSOS.
 - **STAR Precision:** The poor 50.9% precision of STARS can be understood in the model that wrongly identifies 133 true GALAXYs and 30 true QSOs as STARS

This GALAXY-STAR confusion (133 one way, 28 the other) is the one greatest source of inaccuracy in the model.

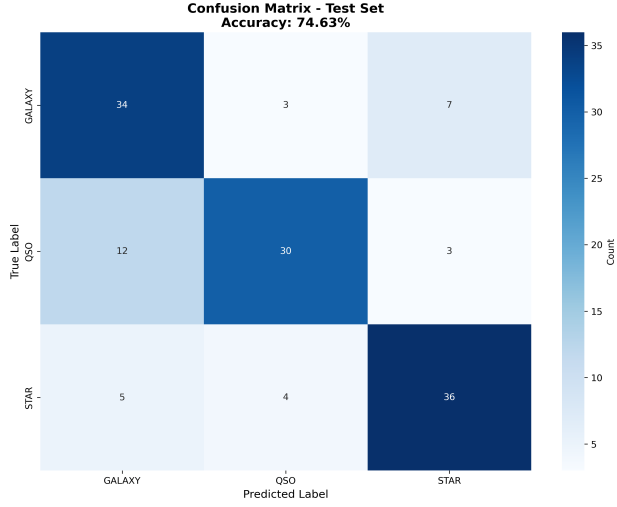


Fig. 4. A Confusion Matrix (Balanced Test Set).

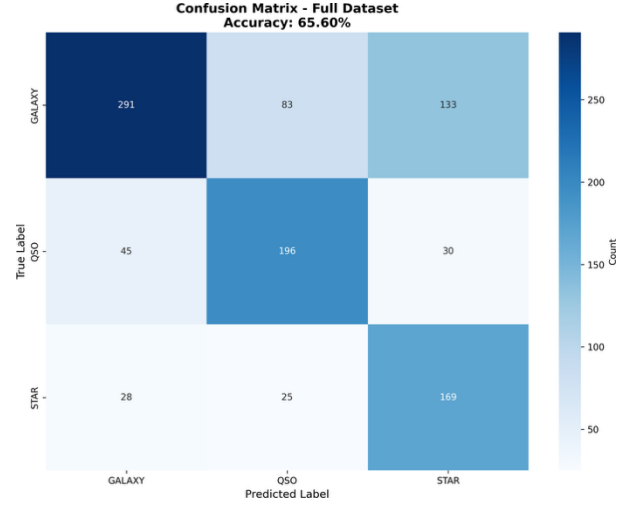


Fig. 4. B Confusion Matrix (Full Imbalanced Dataset).

4.5 Comparison with Baseline Model

Lastly, in order to test the methodology effectiveness, the final optimized model was compared with a baseline SVM model. The simple set up without enhanced feature engineering and hyperparameter optimization was used as the baseline. Table IV makes the balance test set accuracy the comparison between the two models, as this would provide the most logical comparison of the classification capabilities of the two models.

Table 4. Final Model vs Baseline model performance.

Model	Overall Accuracy (Balanced Test Set)
Baseline Model	57.46%
Final Optimized Model	74.63%
Improvement	+17.17 percentage points

The conclusion is straightforward: the last pipeline based on massive feature engineering and hyperparameter optimization makes a strong +17.17 percentage point difference in accuracy compared to the baseline. This significant improvement proves that the stepwise procedure of the features selection and model optimization was very productive.

4.6 Novelty of the Proposed Approach

The novelty of this study lies in the implementation of a multi-domain feature engineering pipeline that specifically targets the morphological ambiguities inherent in celestial object classification without the computational overhead of deep learning. This is in contrast to traditional baseline models that uses raw

intensities of pixels or simple descriptors of shape, our model introduces a 61-dimensional feature space. This space uniquely combines:

- **Structural profiles:** This measures brightness radially at the centroid.
- **Texture analysis:** Measurement of surface properties using Local Binary Patterns (LBP).
- **Log-stabilized Hu Moments:** Numerically and scale invariant rotation descriptors of small and weak objects

With the combination of these unique domains together with the use of Mutual Information to select the features, the model developed is a compact and highly sophisticated. This model is not only feasible in resource-limited settings but also demonstrates high discriminatory ability.

Measurable Performance Advantage: The resulting optimized pipeline generates a large and a quantifiable benefit as performance advantage over the baseline model:

Quantitative Improvement: The final optimized model had an optimized balance-test-set accuracy of 74.63%, which is a significant 17.17% higher than the baseline model of 57.46%.

Generalization and Robustness: The absence of overfitting is one of the performance attributes. The negligible difference in training accuracy (74.44%) and test accuracy (74.63) shows that the process of feature selection has left impressions on generalized astronomical features and not noises.

Precision in Majority Classes: The model achieved a high precision of 79.9% on Galaxies in real-world, imbalanced cases (1,000 samples), ensuring that objects classified as such are highly likely to be accurate, a critical requirement for large-scale cataloging.

Computational Efficiency: The SVM pipeline has high inference (about 0.01ms/sample) compared to the intensive use of CNN-based methods that demand a large number of GPUs, therefore, it can be used on large-scale datasets such as the SDSS.

The model exhibits a systematic bias toward stellar classification because distant galaxies and stars converge into a similar point-like morphology in single-band optical imaging. This shared compactness makes them nearly indistinguishable using shape features alone. This limitation cannot be overcome by the improvement of the algorithms only since the required discriminative information is not provided in the data.

5 Societal Impact and Practical Applications

Modern telescopes record millions of images every night which cannot be analyzed manually by astronomers. Our classification system can be useful to help with this problem. The main advantage of our solution is that it can run on any normal computer. This is because our solution is an SVM pipeline and does not require expensive GPUs like those used by deep learning techniques. Speed is relevant to astronomy. Once telescopes have picked up on some points of interest, researchers

require quick classification with which they can determine the follow-up observations worthy of their attention. Telescope time is valuable and costly. Our solution assists scientists with the efficient processing of information. The challenges we addressed: imbalanced datasets and similar objects they can be found in other application domains as well. These can be considered common challenges. Similar challenges also arise in other domains, such as medical imaging and large-scale data analysis, where class imbalance and visually similar classes are common. Looking forward, with future surveys such as the Rubin Observatory, there will be discoveries of billions of objects. Automated systems, like our classification tool, will play a crucial role in being able to handle this volume of data.

6 Limitations and Future Work

6.1 Key Limitations

The model performed satisfactorily in terms of accuracy but has inherent limitations that must be acknowledged.

1. *Morphological Ambiguity*: One of the primary challenges arises from morphological ambiguity in single-band optical imagery. Celestial bodies often appear visually similar when observed using single-color images. The confusion matrix revealed that 133 galaxies were misclassified as stars, not due to incorrect model behavior, but because distant galaxies appear point-like and visually indistinguishable from stars. The same phenomenon is also observed in the term quasi-stellar objects, which refers to the appearance of the quasar, which looks like a star in a visual image. This ambiguity results in unavoidable precision and recall trade-offs. The classifier becomes conservative in making predictions of galaxies, since actual galaxies are not always found to have extended morphology. As a result, the model will over-predict stars, as stars and distant galaxies are both compact. The limitation cannot be overcome by the improvement of the algorithms only since the required discriminative information is not provided in the data.
2. *Data Constraints*: The research was based on single-band (r-band) images only, and the rest of the four SDSS filters were not used. Each celestial object exhibits distinct color characteristics that can help resolve classification ambiguity; however, such information is unavailable when using single-band data. This limitation required under sampling of other classes and limited the model to being exposed to the full range of astronomical morphology. Consequently, there was a likelihood of underrepresenting uncommon structures like edge-on galaxies and interacting systems.

6.2 Future Research Directions

Based on the findings of this study, several promising directions can be pursued to improve classification performance and address current limitations.

1. **Introduction of Multi-Band Imaging**: An extension of this study would be to use all five SDSS photometric bands (u, g, r, i, z) rather than use just the single-band (r-band). Multi-Band Imaging provides color information that complements morphological features.

Celestial bodies have typical colors that are dependent on the physical features like temperature and the stellar population. Hot stars (blue) contrast with red (cool) stars, in that the spiral galaxies have frequently been found to have younger, blue populations of stars, and the elliptical galaxies are redder as they have older stars. Quasar is characterized by excess of ultraviolet. Filter ratio indices of color like $(g - r)$ and $(u - g)$, introduce an additional discriminative dimension.

Ambiguous objects in terms of their shape may be separated by information of color. For example, a compact blue object can represent a star or a quasar, whereas a compact red object can represent a star or a galaxy. Implementation is straightforward: process multiple images per object, compute color features, and allow feature selection to determine relevance.

Expected Impact: Multi-band photometry may potentially boost the classification accuracy to between low and mid 80 percent, especially in star-galaxy, and this directly deals with morphological ambiguity.

2. **Advanced Feature Engineering:** Future work should incorporate well-established astronomical descriptors that were not utilized in this study, including:
 - **Se'rsic profiles:** Gradually qualify the decrease of galaxy brightness with radius, and also effectively distinguish galaxy type.
 - **Concentration indices:** Ratios of light in inner versus outer regions that distinguish compact from extended sources.
 - **Surface brightness profiles:** Measure structural variations between elliptical galaxies, spiral disks and quasar cores.
 - **Color gradients:** Characterize gradients in color between core and periphery that reflect the distributions of their stellar populations.

These physically inspired features can store differences that cannot be represented by properties that are purely data based.

3. **Incorporation of Spectroscopic Data:** Spectroscopy offers a clear-cut solution to the problem of ambiguity in classification. Spectroscopy allows recording of numerous measurements in hundreds of wavelengths (as opposed to photometry which provides few measurements), which allows:
 - Determination of absorption lines in stellar atmospheres.
 - Observation of emission lines of stellar populations of galaxies.
 - Identification of widespread emission lines in the case of quasars.
 - Direct redshift measurement, differentiating the stars around a galaxy and the distant galaxies and quasars.
4. **Cross-Survey Validation and Generalization:** Implementing the suggested pipeline on other datasets like Pan-STARRS and the Dark Energy Survey would allow assessing the generalizability beyond of SDSS. It would indicate data specific biases and it would assist to develop more general methods of classification.

5. **Active Learning for Resource Optimization:** Active learning measures can be implemented to find objects whose further spectroscopic observations would be most beneficial. By prioritizing uncertain cases, telescope time can be used more efficiently while maximizing performance gains.
6. **Integration of Deep Learning:** Another alternative approach might be a hybrid model instead of the SVM classifier. Convolutional neural networks (CNNs) may serve as feature extractors whereby they produce embeddings that are fed into the SVM. Such a method has a smaller number of training samples than end-to-end deep learning, and may be able to learn intricate morphological patterns.
7. **Dataset Expansion:** Expanding the dataset, particularly for stars and quasars, would improve the model's ability to generalize across diverse morphologies. Additionally, rarer and less frequent structures would also be incorporated in a bigger dataset and make it more robust.

7 Open Challenges

There are several broader challenges that remain in astronomical classification:

1. **Extreme class imbalance:** For real surveys, there are many more galaxies than stars or quasars, and this requires better training methods.
2. **Rare object detection:** The detection of scientifically interesting outliers, such as gravitational lenses and exotic transients, is a challenging problem.
3. **Uncertainty quantification:** Calibrated confidence estimates necessary to facilitate risk-aware follow up decisions should be incorporated through these models.
4. **Temporal information:** This addition of multi-epoch variability also introduces new challenges in handling irregular time-series data.
5. **Real-time classification:** Future surveys require near real-time processing without sacrificing accuracy. The challenges support the topicality of this work and create great possibilities of the further improvement and innovation.

8 Conclusion

This study has developed and tested a complete machine learning pipeline to categorize astronomical objects with an overall accuracy of 74.6% on balanced data 17% above the baseline model. The paper has shown that morphological classification can be effectively and accessibly solved using SVMs with careful feature engineering. But however, our analysis shows, single-band methods of morphology have inherent constraints. A severe morphological ambiguity between galaxies, quasars and stars has precision-recall trade-offs, which cannot be removed by a more sophisticated algorithm. The performance of single-band images is limited by intrinsic limitations introduced by its information capacity. The given work is functional in its classification system as well as provides a valuable perspective to the challenges and opportunities of the context of astronomical classification. It is not very long before the nature of astronomical surveys is going to expand to such extent that the solutions we have arrived to and

the lessons we have learned will be used to make the next generation of classifiers aware of what they are capable of in tackling the problems of the new age of modern astronomy.

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